DataSynth

Enterprise Synthetic Data Platform

High-Performance Generation for Accounting, Audit & ML

Rust core • Python 1.0 • REST/gRPC API • Desktop UI • Helm/Docker

Executive Overview

Version 0.5.0

Deterministic, privacy-preserving synthetic data generation
for enterprise finance, compliance testing, and machine learning.

Built with Rust & Python • 15 modular crates • 100 K+ entries/sec

Open-source (Apache-2.0) • Commercial license available

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1 Executive Summary

DataSynth is a high-performance synthetic data platform purpose-built for enterprise accounting, audit analytics, and machine learning. Written in Rust for maximum throughput and memory safety — with a full Python wrapper for data-science workflows — it generates realistic, internally coherent financial data that satisfies the statistical and structural properties of real-world enterprise resource planning (ERP) systems.

Why DataSynth?

- **No real data required** — eliminates privacy, regulatory, and procurement barriers to analytics development.
- **Full auditability** — deterministic, seeded generation means every dataset is perfectly reproducible.
- **ML-ready from day one** — ground-truth labels for fraud, anomalies, and data quality ship alongside every record.
- **Domain depth** — covers the full accounting lifecycle: journal entries, document flows, subledgers, FX, intercompany, period close, and banking/AML.
- **Empirically grounded** — statistical distributions for journal entry line items, amount patterns, and temporal volumes are calibrated against empirical research conducted on real-world enterprise datasets, ensuring synthetic output mirrors the structural properties of production ERP data.

1.1 At a Glance

100 K+ entries/sec (single thread)	**50+** ACFE-aligned fraud schemes	**10** industry presets
15 modular Rust crates	**40+** red flag patterns	**4** privacy levels

2 Platform Capabilities

2.1 Enterprise Accounting Simulation

DataSynth generates a complete, internally consistent accounting universe:

Domain	Capabilities
Journal Entries	Balanced debits/credits, Benford-compliant amounts, configurable line-item distributions, SAP ACDOCA format export.
Master Data	Vendors, customers, materials, fixed assets, employees with hierarchies, payment terms, credit ratings, and intercompany flags.
Document Flows	Full Procure-to-Pay (PO → GR → Invoice → Payment) and Order-to-Cash (SO → Delivery → Invoice → Receipt) with three-way match validation.
Intercompany	Matched IC journal entry pairs, transfer pricing (Cost-Plus, Resale-Minus, CUP), and consolidation elimination entries.
Subledgers	AR/AP open items and aging, fixed asset register with IRS MACRS depreciation tables (3–20 yr recovery periods), inventory positions and movements, GL-to-subledger reconciliation.
FX & Translation	Ornstein–Uhlenbeck exchange rate process, multi-currency trial balance translation (current-rate, temporal, monetary/non-monetary methods with account-type classification), currency translation adjustment entries.
Period Close	Month-end accruals, depreciation runs, year-end closing entries, fiscal period status tracking.

2.2 Banking, KYC & AML

A dedicated banking module generates realistic transaction data for anti-money-laundering testing:

- **Customer personas**: Retail, Business, Trust profiles with full KYC envelopes (declared turnover, source of funds, geographic exposure, cash intensity).
- **AML typologies**: Structuring, funnel accounts, layering schemes, money mule networks, round-tripping, and adversarial spoofing for robustness testing.
- **Ground-truth labels**: Entity-level risk classifications, transaction-level labels, and investigation narratives.

2.3 Audit Simulation

Generates ISA-compliant audit artifacts:

- Engagement metadata with materiality thresholds (ISA 320).
- Workpapers per ISA 230, evidence per ISA 500.
- Risk assessments (ISA 315/330), findings (ISA 265), and professional judgment documentation (ISA 200).

2.4 COSO 2013 Internal Control Framework

Full integration with the COSO Internal Control-Integrated Framework:

- **5 Components**: Control Environment, Risk Assessment, Control Activities, Information & Communication, Monitoring Activities.
- **17 Principles**: Complete principle coverage with control mappings.
- **Control Scopes**: Entity-level, transaction-level, IT general, and IT application controls.
- **Maturity Levels**: 6-level model (Non-Existent through Optimized).
- **Export**: `coso_control_mapping.csv` with principle-level granularity.

2.5 Accounting & Audit Standards

Comprehensive standards support via the `datasynth-standards` crate:

Category	Standards Supported
Accounting (US GAAP)	ASC 606 (Revenue), ASC 842 (Leases — 5 bright-line classification tests), ASC 820 (Fair Value), ASC 360 (Impairment)
Accounting (IFRS)	IFRS 15 (Revenue), IFRS 16 (Leases — enhanced ROU measurement), IFRS 13 (Fair Value), IAS 36 (Impairment)
Audit Standards	34 ISA standards (ISA 200–720), 19+ PCAOB standards with ISA mapping
Regulatory	SOX Section 302 (Certifications), SOX Section 404 (ICFR Assessment), Deficiency Matrix (MW/SD classification)

Supports dual reporting mode (US GAAP + IFRS) with automatic framework difference reconciliation.

2.6 Process Mining (OCEL 2.0 & XES 2.0)

Object-Centric Event Logs track many-to-many relationships between business objects (orders, invoices, payments) and activities — enabling conformance checking and process variant analysis.

- **OCEL 2.0**: JSON and XML export with objects, events, and object relationships.
- **XES 2.0**: IEEE standard format for ProM, Celonis, Disco, and pm4py.
- **Reference Models**: Canonical P2P (9 activities), O2C (10 activities), and R2R (11 activities) process models for conformance checking.
- **17 R2R/A2R Activities**: GL posting, FX revaluation, period close, depreciation, consolidation.
- **Deterministic UUIDs**: FNV-1a hash-based IDs with type discriminators for reproducible event logs.

3 AI-Augmented Generation

New in v0.5.0 — three opt-in AI backends that enhance the rule-based core without sacrificing determinism or reproducibility.

3.1 LLM-Augmented Metadata Enrichment

An `LlmProvider` trait integrates external language models for contextual metadata enrichment while preserving deterministic fallback:

- **Provider trait**: `MockProvider` (deterministic, default) and `HttpProvider` (OpenAI, Anthropic, or custom API) implementations.
- **Enrichment targets**: Vendor names, transaction descriptions, anomaly explanations, and memo fields.
- **Natural language config**: Generate YAML configurations from plain English descriptions.
- **Response caching**: LRU cache with deterministic fallback ensures reproducibility even with stochastic providers.

3.2 Diffusion Model Integration

A statistical diffusion backend learns distribution shapes from fingerprints and blends them with rule-based outputs:

- **Noise schedules**: Cosine, linear, and sigmoid with configurable step count.
- **Hybrid mode**: Weighted blend of rule-based and diffusion-generated values (`diffusion_weight` 0.0–1.0).
- **Training pipeline**: `DiffusionTrainer` with epoch count, batch size, and learning rate configuration.

3.3 Causal & Counterfactual Generation

Structural causal models enable "what-if" scenario generation grounded in do-calculus:

- **CausalGraph**: DAG specification of causal relationships between entities (e.g., vendor risk → invoice amount → approval level).
- **StructuralCausalModel**: Functional causal model with noise terms and topological evaluation.
- **Interventions**: Do-calculus interventions for hypothetical scenarios ("what if vendor risk doubled?").
- **Counterfactuals**: Generate alternative versions of existing records under modified assumptions.
- **Templates**: Pre-built causal graphs for fraud detection, revenue impact analysis, and supply chain modeling.

4 Machine Learning & Analytics

4.1 ACFE-Aligned Fraud Taxonomy

DataSynth implements the ACFE (Association of Certified Fraud Examiners) "Report to the Nations" taxonomy with calibrated statistics:

ACFE Category	Prevalence	Scheme Types
Asset Misappropriation	86%	20+ cash schemes: skimming, larceny, shell company, ghost employee, billing fraud, expense reimbursement
Corruption	33%	Conflicts of interest, bribery, kickbacks, bid rigging, economic extortion
Financial Statement	10%	Revenue manipulation, expense timing, concealed liabilities, improper disclosures

Calibration: $117k median loss, 12-month median duration, detection method distribution (42% tips, 16% internal audit, 12% management review).

4.2 Advanced Fraud Modeling

- **Collusion & Conspiracy**: 9 ring types (EmployeePair, DepartmentRing, EmployeeVendor, VendorRing), 6 conspirator roles, defection modeling based on detection risk.
- **Management Override**: Executive-level fraud with override techniques, Fraud Triangle modeling (Pressure, Opportunity, Rationalization), concealment strategies.
- **Red Flag Patterns**: 40+ patterns with Bayesian probabilities, 6 categories (Vendor, Transaction, Timing, Approval, Document, Behavioral), realistic false positive rates.
- **Industry-Specific**: Manufacturing (yield manipulation, phantom production), Retail (sweethearting, refund fraud), Healthcare (upcoding, unbundling, phantom billing).

Anomaly injection is **entity-aware**: injection rates are adjusted per vendor, employee, and account based on contextual risk factors (e.g., new vendors, high-value accounts, employees with recent access changes). Every injected anomaly carries a `LabeledAnomaly` record with full metadata, enabling supervised and semi-supervised learning pipelines without manual labeling effort.

4.3 Data Quality Variations

Realistic data imperfections for training data-quality ML models:

- **Missing values**: MCAR, MAR, MNAR, and systematic patterns.
- **Format variations**: Date, amount, and identifier format diversity across regional conventions.
- **Duplicates**: Exact, near, and fuzzy duplicates.
- **Typos**: Keyboard-aware substitution, OCR errors, homophones.
- **Encoding issues**: Mojibake, BOM artifacts, HTML entity corruption.

4.4 Enterprise Interconnectivity

Realistic enterprise network modeling with configurable relationships:

- **Multi-Tier Vendor Networks**: 3-tier supply chain (Tier1/Tier2/Tier3), vendor clusters (ReliableStrategic 20%, StandardOperational 50%, Transactional 25%, Problematic 5%), lifecycle stages, quality scores, dependency analysis.
- **Customer Value Segmentation**: Enterprise (40% revenue/5% customers), MidMarket (35%/20%), SMB (20%/50%), Consumer (5%/25%), lifecycle stages, referral networks, corporate hierarchies.
- **Entity Relationship Graph**: 16 entity types, 26 relationship types, composite strength calculation from volume, count, duration, recency, and mutual connections.
- **Cross-Process Links**: P2P↔O2C linkage via inventory (GoodsReceipt→Delivery), intercompany bilateral links.

4.5 Pattern & Process Drift

Comprehensive drift modeling for realistic temporal evolution:

- **Organizational Events**: Acquisition (1.35x volume, 5% integration errors), divestiture, reorganization, leadership change, workforce reduction, merger.
- **Process Evolution**: Approval workflow changes, S-curve automation rollout, policy changes, control enhancements.
- **Technology Transitions**: ERP migration phases with error rate and processing time multipliers, duplicate/missing/format issues.
- **Behavioral Drift**: Vendor payment term extensions, customer delays during downturns, employee learning curves, year-end intensity.
- **Market Drift**: Economic cycles (sinusoidal, asymmetric, mean-reverting), industry-specific cycles, commodity drift, price shocks.
- **Ground Truth**: `LabeledDriftEvent` with event type, magnitude, and detection difficulty (Easy/Medium/Hard) for drift detection ML.

4.6 Graph & Network Export

Supported Graph Formats

PyTorch Geometric	`.pt` files with node features, edge index, edge attributes, labels, and train/val/test masks.
Neo4j	CSV node/edge files with Cypher import scripts.
DGL	Deep Graph Library format for GNN training.
RustGraph	Native Rust graph format with adjacency lists.
RustGraph Hyper	Multi-layer hypergraph export (entity, temporal, process layers with cross-layer edges).

All five exporters share a unified `CommonExportConfig` for consistent feature engineering, label selection, and train/val/test mask settings. Computed features include temporal signals (weekday, period-end flags), amount signals (log-amount, Benford probability, round-number flag), structural signals (line count, unique accounts), and one-hot categorical encodings.

5 Privacy & Fingerprinting

5.1 Privacy-Preserving Fingerprint Extraction

DataSynth can extract a statistical *fingerprint* from real data and synthesize new data that matches its properties — without ever copying individual records.

5.2 Privacy Levels

Level	Epsilon (ε)	k-Anon	Outlier %	Use Case
Minimal	5.0	3	99%	Low privacy, high utility
Standard	1.0	5	95%	Balanced (default)
High	0.5	10	90%	Sensitive environments
Maximum	0.1	20	85%	Maximum privacy guarantees

The privacy engine combines differential privacy (Laplace and Gaussian mechanisms), k-anonymity with rare-value suppression, and outlier winsorization. A full privacy audit trail records every decision and the cumulative epsilon budget spent.

5.3 Federated Fingerprinting & Certificates

New in v0.5.0 — distributed fingerprint extraction and cryptographic provenance:

- **Federated Coordinator**: Aggregate fingerprints from distributed data sources without centralizing raw data.
- **Synthetic Data Certificates**: `SyntheticDataCertificate` with cryptographic proof of differential privacy guarantees, generation metadata, and validation workflow.
- **Pareto Frontier**: Automated exploration of optimal privacy–utility trade-offs across epsilon values.

5.4 Formal DP Composition

New in v0.5.0 — advanced privacy accounting beyond naive ε-summation:

- **Rényi DP**: RDP curve tracking at α values 2–128 with conversion to (ε,δ)-DP.
- **zCDP**: Zero-Concentrated DP with additive ρ composition and tighter bounds.
- **Budget Manager**: Global budget tracking across extraction runs with composition-aware enforcement.
- **Privacy Evaluation**: Post-generation membership inference attack (MIA) testing, linkage attack assessment, and NIST SP 800-226 alignment.

5.5 Fidelity Evaluation

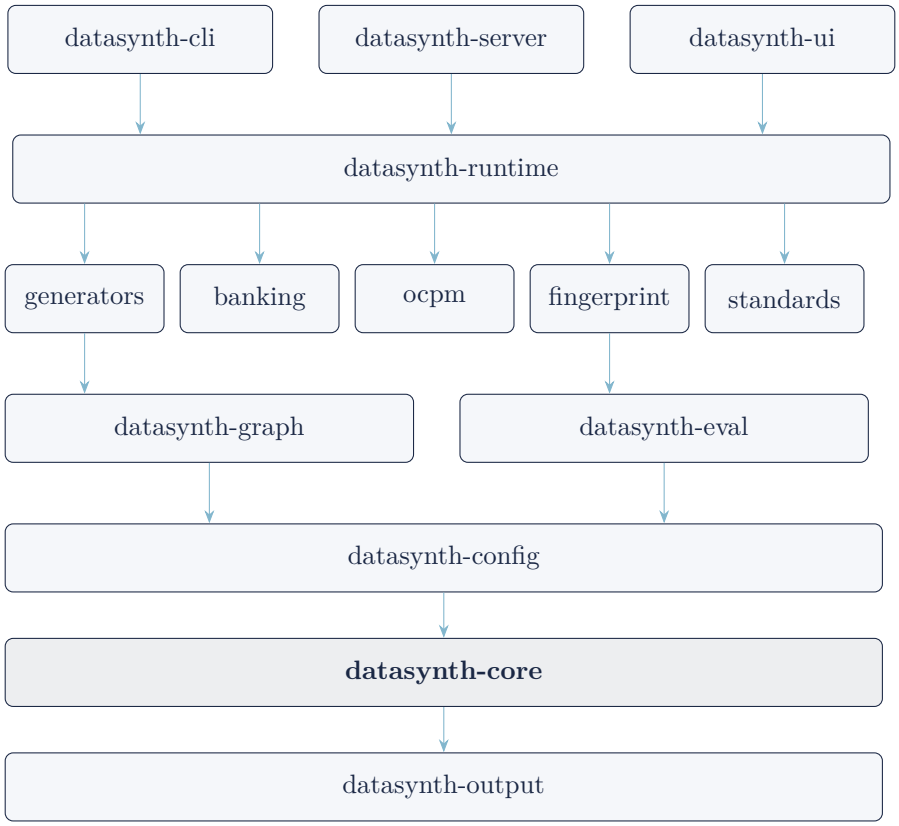
Per-column statistical distance metrics quantify how well synthetic data matches the source fingerprint:

Metric	Description
KS Statistic	Kolmogorov-Smirnov two-sample test statistic (sup of CDF difference).
Wasserstein-1	Earth Mover's Distance via piecewise-linear inverse CDF integration over 9 percentile knots.
JS Divergence	Jensen-Shannon divergence from percentile-bin PMFs, bounded by $\ln 2$.

Distribution CDFs are computed analytically for Normal, LogNormal, Gamma (regularized incomplete gamma via Lanczos + Lentz continued fraction), Pareto, PointMass, and Mixture distributions.

6 Architecture

6.1 Layered Crate Architecture



6.2 Production-Grade Infrastructure

Capability	Details
Deterministic Output	ChaCha8 PRNG with configurable seed; identical seed ⇒ identical dataset.
Financial Precision	`rust_decimal` throughout; no IEEE 754 floating-point artifacts. Decimals serialized as strings.
Resource Guards	Unified CPU, memory, and disk monitoring with automatic throttling and graceful degradation (Normal → Reduced → Minimal → Emergency).
Collision-Free IDs	FNV-1a hash-based UUID factory with generator-type discriminators prevents document ID collisions across parallel generators.
Streaming Output	CSV, JSON, NDJSON, and Apache Parquet (15-column Arrow schema, Zstd compression) streaming sinks with backpressure strategies (Block, DropOldest, DropNewest, Buffer).
API Layer	REST + gRPC + WebSocket with Argon2id API-key auth, JWT/OIDC bearer tokens (RS256, Keycloak/Auth0/Entra ID), RBAC (Admin/Operator/Viewer), sliding-window rate limiting (in-memory or Redis), TLS (rustls), async job queue, security headers, and structured audit logging.
Desktop UI	Tauri + SvelteKit cross-platform application with 15+ configuration pages and real-time streaming viewer.
Plugin SDK	`GeneratorPlugin`, `SinkPlugin`, and `TransformPlugin` traits with thread-safe registry for custom extensions.
Webhooks	Fire-and-forget event dispatch (RunStarted, RunCompleted, RunFailed, GateViolation) with HMAC signing and configurable retry.
Data Lineage	Per-file SHA-256 checksums, `LineageGraph` tracking config→generator→output relationships, W3C PROV-JSON export, CLI `verify` command.
Python Wrapper	`datasynth-py` 1.0 with blueprints, pandas/polars integration, async client, and WebSocket streaming support.

7 Industry Presets

DataSynth ships with ten industry-specific configuration presets, each tuned for realistic business process weightings, chart-of-accounts structures, and regional multi-company setups.

Each preset supports three complexity tiers: **Small** (∼100 GL accounts), **Medium** (∼400 accounts), and **Large** (∼2 500 accounts).

8 Statistical Distribution Framework

DataSynth includes an advanced statistical distribution framework for generating empirically grounded synthetic data:
All distributions include validation tests: Benford's Law MAD, Anderson-Darling goodness-of-fit, chi-squared, and correlation matrix verification.

Industry	Key Weight	Characteristics
Manufacturing	40% P2P	14 transaction types, multi-level BOM, production routings, work centers, JIT, quality framework.
Retail	50% O2C	12 transaction types, 7 store types, shrinkage modeling, loss prevention, markdown patterns.
Healthcare	15% H2R	15 transaction types, payer mix (Medicare/Medicaid/Commercial), ICD-10/CPT/DRG coding, HIPAA/Stark compliance.
Financial Services	40% R2R	Report-heavy, regulatory compliance, intercompany structures, loan loss provisions.
Technology	15% H2R	SaaS revenue recognition, R&D capitalization, deferred revenue.
Professional Svcs.	—	Time-based billing, project accounting, trust accounting.
Energy	—	Capital-intensive, long-lived assets.
Transportation	—	Fleet management, route-based costing.
Real Estate	—	Property portfolios, lease accounting.
Telecommunications	—	Subscription revenue, network assets.

Capability	Details
Mixture Models	Gaussian and Log-Normal mixtures with weighted components ("routine", "significant", "major" transactions).
Copula Correlation	Gaussian, Clayton, Gumbel, Frank, Student-t copulas for cross-field dependencies.
Distribution Types	Pareto (heavy-tailed), Weibull (time-to-event), Beta (proportions), Zero-inflated (excess zeros).
Benford's Law	First and second-digit compliance, round number bias, threshold clustering.
Regime Changes	Economic cycles, acquisition/divestiture effects, recession modeling.
Industry Profiles	Pre-configured profiles for Retail, Manufacturing, Financial Services.

9 Use Cases

Primary Use Cases

- **Fraud Detection ML** — Train and validate models on labeled fraud typologies with realistic base-rate imbalance.
- **Graph Neural Networks** — Export transaction graphs in PyTorch Geometric, Neo4j, DGL, or RustGraph format — including multi-layer hypergraphs — with pre-computed features and train/val/test splits.
- **AML & KYC Testing** — Generate banking transaction data with structuring, layering, and mule patterns for compliance system validation.
- **Audit Analytics** — Produce ISA-compliant audit artifacts and anomaly-injected financial data for analytics tool development.
- **ERP Integration Testing** — Generate SAP ACDOCA-format data with full document chains for system migration and integration testing.
- **Process Mining** — OCEL 2.0 event logs for process discovery, conformance checking,

10 Evaluation & Auto-Tuning

DataSynth includes a built-in evaluation framework that measures synthetic data quality across four dimensions:

1. **Statistical Fidelity** — KS statistic, Wasserstein-1 (Earth Mover's Distance), Jensen-Shannon divergence, Benford's Law MAD, Anderson-Darling goodness-of-fit.
2. **Coherence** — Balance-sheet validation, intercompany matching, document chain integrity, standards compliance.
3. **Data Quality** — Completeness, consistency, duplicate rates, format correctness, uniqueness.
4. **ML Readiness** — Feature distributions, label quality, graph structure, train/val/test split balance.
5. **Drift Detection** — Hellinger distance, Population Stability Index (PSI), rolling window analysis, detection delay metrics.

10.1 ML Benchmarks

Pre-configured benchmarks aligned with ACFE statistics and industry verticals:

- **ACFE-Calibrated**: General fraud (1k), collusion-focused (5k), management override (2k).
- **Industry-Specific**: Manufacturing fraud (5k), retail fraud (10k), healthcare fraud (5k), technology fraud (3k), financial services fraud (5k).
- **Cost-Sensitive**: Asymmetric cost matrices for realistic fraud detection evaluation.

An **auto-tuning engine** analyses evaluation results and produces prioritized configuration patches with expected improvement estimates — closing the loop between generation and validation.

10.2 Quality Gate Engine

New in v0.5.0 — configurable pass/fail thresholds that can block releases of low-quality synthetic data:

- **8 metrics**: Benford MAD, balance coherence, document chain completeness, duplicate rate, missing rate, distribution fit, correlation accuracy, anomaly precision.
- **Built-in profiles**: `strict`, `default`, `lenient` — or define custom per-metric thresholds.
- **CLI integration**: `-quality-gate <profile>` with exit code 2 on failure for CI/CD pipelines.

11 Compliance & Governance

11.1 EU AI Act

DataSynth implements Article 50 (synthetic content marking) and Article 10 (data governance) requirements:

- **Content Credentials**: `SyntheticContentMarker` creates machine-readable credentials (generator name, version, timestamp, config hash) in embedded, sidecar, or both formats.
- **Data Governance Report**: `DataGovernanceReport` with `BiasAssessment` documenting data provenance, quality metrics, and bias evaluation for regulatory submission.

11.2 Regulatory Compliance Documentation

Pre-built compliance guides with self-assessment templates:

Framework	Coverage
EU AI Act	Article 50 content marking, Article 10 governance
NIST AI RMF	Self-assessment across MAP, MEASURE, MANAGE, GOVERN
GDPR	Article 30 records, DPIA guidance, data minimization
SOC 2 Type II	Readiness across 5 Trust Service Criteria
ISO 27001:2022	Annex A alignment (11 implemented, 6 partial)

12 DevOps & Deployment

12.1 CI/CD Pipeline

A 7-job GitHub Actions pipeline ensures release quality:

1. **fmt**: `cargo fmt -check`
2. **clippy**: Lint with `-D warnings`
3. **test**: Cross-platform matrix (Ubuntu, macOS, Windows)
4. **msrv**: Minimum Supported Rust Version validation (1.88)
5. **security**: `cargo deny` + `cargo audit`
6. **coverage**: `cargo-llvm-cov` with Codecov
7. **benchmarks**: Criterion regression checks on PRs

12.2 Deployment Options

Target	Details
Docker	Multi-stage builds with cargo-chef caching; distroless runtime images for server and CLI.
Docker Compose	Server + Prometheus + Grafana stack with auto-provisioned datasources.
Kubernetes	Helm chart with HPA (2–10 replicas), PDB, rolling updates, optional Redis subchart, Prometheus ServiceMonitor, and ConfigMap/Secret templates.
Bare Metal	SystemD service with security hardening (NoNewPrivileges, ProtectSystem, MemoryMax=4G, CPUQuota=200%).
Release Pipeline	Pre-built binaries for 5 platforms, Docker image to GHCR (amd64 + arm64), Trivy container scanning.

12.3 Ecosystem Integrations

Python-native integrations for popular data engineering tools:

- **Apache Airflow**: `DataSynthOperator`, `DataSynthSensor`, and `DataSynthValidateOperator` for orchestrated generation.
- **dbt**: `DbtSourceGenerator` for source YAML and seed files.
- **MLflow**: `DataSynthMlflowTracker` for experiment tracking with quality metrics.
- **Apache Spark**: `DataSynthSparkReader` for direct DataFrame loading from DataSynth output.

13 Getting Started

13.1 Quick Start (CLI)

```
# Generate with demo preset
datasynth-data generate -demo -output ./output

# Create an industry-specific config
datasynth-data init -industry manufacturing -complexity medium -o config.yaml

# Generate from config
datasynth-data generate -config config.yaml -output ./output
```

13.2 Quick Start (Python)

```
from datasynth_py import DataSynth
from datasynth_py.config import blueprints

config = blueprints.retail_small(companies=4, transactions=10000)
synth = DataSynth()
result = synth.generate(config=config)
```

13.3 Server Mode

```
# Start REST/gRPC server with 4 worker threads
cargo run -p datasynth-server - -port 3000 -worker-threads 4
```

Links & Resources

Repository	https://github.com/ey-asu-rnd/SyntheticData
Documentation	https://ey-asu-rnd.github.io/SyntheticData/
Crates.io	https://crates.io/crates/datasynth-core
PyPI	https://pypi.org/project/datasynth-py/

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